Lesson Learned Card 经验教训卡

Customer plant客户工厂	Kelier	**Supplier** 供应商	AVO Anhui/Tianjin	**LLC Type** LLC类型	Detection☐ 探测 / Logistics☐ 物流
Claim reference投诉编号	20250929	**Author** 审核者	Tim		Occurrence☐ 发生 / Process☐ 制程
Date of incident发生日期	2025/9/28	**Date of LLC** LLC日期			Management☐ 管理 / Product☐ 产品

1. WHAT WAS THE PROBLEM? (5W + 2H)问题描述（5W2H）

What happened ? 发生了什么？
Standard:: 25±1
Actual size:: 18.74
Why it is a problem ? 为什么这是一个问题？
影响使用，碳刷无法正常工作
Affects usage, carbon brushes cannot function properly
When did it happen ? 何时发生的？
45928
Who has detected it ? 何人发现的？
OAperator
Where has been detected ? 何处发现的？
Assembly line
How has it been detected ? 如何发现的？
visual inspection
How many ? 多少数量？
1/4000

2. WHAT IS THE CAUSE? 原因

Explain the root cause that led to the occurrence or the non detection of the problem according to the type of LLC.

Abstract of FTA & 5 Why's (Occurrence or non detection).
Why and How the Factor is linked the failure.
It is not the defect description

跟踪LLC 类型描述发生或流出的根本原因
FTA 及5W 的概要（发生或流出）
为什么及怎么样的因子关联至该不良
该处不是填写不良描述

3. DESCRIPTION OF THE APPLIED MODIFICATION 对变更的描述

BEFORE 之前

Add a picture or a descriptive graph which explains the conditions that led to the problem 添加图片或图示解释问题发生的状态

Situation prior to countermeasures application
Process:
Photographs of machine, control device

Method:
Flow chart
Test method - conditions
Product:
Component design

在对策之前的原始状态
制程：
机器设备，控制器的照片
方法：
流程图
测试方法，测试条件
产品：
产品设计

AFTER 之后

Add a picture or a descriptive graph which explains the countermeasure on the part/process 添加图片或图示解释问题对策展开后的产品/制程
IMPORTANT : "AFTER" picture or graph should be SIMILAR (same point of view, same size) to "BEFORE" picture
重点：‘之后’的图片或图示必须和‘之前’的类似（相同的视图，相同大小）

OK: Picture is same size, same angle
OK：图片是相同尺寸相同角度

NOK : Picture is not the same size, not same angle
NG：图片不是相同尺寸相同角度

Situation after countermeasures application
Process:
Photographs of machine, control device
Method:
Flow chart
Test method - conditions
Product:
Component design

在对策展开之后的状态
制程：
机器设备，控制器的照片
方法：
流程图
测试方法，测试条件
产品：
产品设计

4) WHAT WE LEARNED? 我们学习到什么？

ERADICATION	CHECKING	SIGNALING
What is the countermeasure that was put in place to avoid recurrence ? 根除 防止再发的对策是什么？	What is the method (who, what, how) for the countermeasures? 检测 对策的方法是什么样的（谁，什么，如何）	What is the countermeasure supposed to do or make or hightlight when detecting a non conformity ? 信号 当探测到不合格品时对策提出了哪些方法或重要提示？
Automatic Air pressure sensor at the stroke inlet must be between 5 and 6 bar 自动空气压力传感器在冲程入口必须介于5和6之间	Who? Operator What? Measure air pressure How? Using pressure gauge Nb 123, every 2 hours 谁？操作员 什么？测量气压 如何？使用压力计Nb123，每2小时	If measurement is out of range, red bulb is lighted and line automatically stopps 如果测量超出范围，红色灯将亮起和产线将自动停止

5. WHAT HAS BEEN MODIFIED ? 有哪些被修改？

a. Proces is modified according to Before/ After sheet 根据前后对照表制程被修改了	b. All documents have been updated and has been validated with the operators 所有文件被更新并被操作者验证	c. Procedures and work instructions 程序手册及作业指导书	d. Control Plan and PFMEA 控制计划及PFMEA	e. Maintenance instructions 保养手册	f. Records (start-up checklists, reaction sheets, preventive maintenance...) 记录（始业点检、反应表单、预防保养等）	g. Visual aids are posted 视觉辅助工具	h. Operators and other functions are trained 操作者及其他部门人员被教育训练	i. Proof of training is avalaible (records,5' meeting) 教育训练的证据	j. Operators can correctly describe what they must do 操作者能正确地描述他们必须做什么	k. Other: 其他	l. Other: 其他

6. POTENTIAL APPLICABILITY 潜在适用

Product 产品 :	*UP28 Brush*
Line / process 产线/制程	*UP28 Brush Line*
Plant 工厂 :	*AVO Tianjin*